

Lab 2 - Agile Backlog & Sprint Simulation (Jira)

PES1UG24CS118 | Warehouse Inventory & Pallet Location Tracker

This planning sheet converts the Lab 1 functional requirements into Epics and User Stories, with priorities, Fibonacci story points and a two-sprint plan. Type these straight into Jira, then capture the required screenshots.

EPIC 1 - Pallet Identification & Scanning

Let operators identify any pallet instantly through barcode/RFID instead of manual entry. (Lab 1: FR-002, UC-01)

ID	User Story (As a / I want / So that)	Priority	Story Pts	Sprint
US-1.1	As a Warehouse Operator, I want scan a pallet's barcode or RFID tag, So that the system identifies the pallet automatically, with no manual typing.	High	3	Sprint 1
US-1.2	As a Warehouse Operator, I want see the scanned pallet's inventory details on screen, So that I can confirm I am handling the right stock.	High	2	Sprint 1
US-1.3	As a Warehouse Operator, I want be stopped when I scan an unknown or invalid tag, So that I never act on a pallet that is not in the system.	Medium	2	Sprint 2

EPIC 2 - Pallet Placement & Storage

Place pallets safely into 3D shelf positions while respecting each rack's weight limit. (Lab 1: FR-001, FR-003, UC-02/03/04)

ID	User Story (As a / I want / So that)	Priority	Story Pts	Sprint
US-2.1	As a Warehouse Operator, I want the system to check a rack's maximum weight before I place a pallet, So that I can never overload a rack and risk a collapse.	High	5	Sprint 1
US-2.2	As a Warehouse Operator, I want the pallet's full warehouse / rack / shelf / slot position saved, So that its exact 3D location is always on record.	High	3	Sprint 1
US-2.3	As a Warehouse Operator, I want a clear warning and a block when a placement would exceed the rack limit, So that I can pick a safe rack instead.	High	3	Sprint 2

EPIC 3 - Stock Movement Tracking

Keep a complete, auditable trail of every pallet movement. (Lab 1: FR-004, UC-05, UC-07)

ID	User Story (As a / I want / So that)	Priority	Story Pts	Sprint
US-3.1	As a Warehouse Operator, I want every pallet movement logged with pallet ID, source, destination, timestamp and my ID, So that all stock movements are fully traceable.	High	3	Sprint 1
US-3.2	As a Logistics Supervisor, I want to view a pallet's full movement history, So that I can audit where a pallet has been.	Medium	2	Sprint 2

EPIC 4 - Inventory Search & Lookup

Find any pallet fast, even under heavy load. (Lab 1: FR-005, NFR-001, UC-06)

ID	User Story (As a / I want / So that)	Priority	Story Pts	Sprint
US-4.1	As a Warehouse Operator, I want to search for a pallet by SKU, barcode or RFID, So that I can quickly find its current shelf coordinates.	High	3	Sprint 1
US-4.2	As a Logistics Supervisor, I want lookups to return in under 100 ms across 100,000 bins, So that searching stays fast even at peak load.	Medium	8	Sprint 2

EPIC 5 - Access Control & Security

Make sure only the right people can change inventory data. (Lab 1: NFR-002)

ID	User Story (As a / I want / So that)	Priority	Story Pts	Sprint
US-5.1	As a Logistics Supervisor, I want inventory and stock-movement actions restricted by user role, So that only authorised staff can change records.	High	5	Sprint 2
US-5.2	As a Warehouse Operator, I want to sign in before performing any action, So that every action is attributed to the person who did it.	Medium	2	Sprint 2

Sprint Plan

Two one-week sprints. Story points chosen from the Fibonacci scale (1, 2, 3, 5, 8, 13...). Velocity is deliberately similar across both sprints so the burndown reads cleanly.

Sprint	Stories included	Points	Total	Sprint goal
Sprint 1	US-1.1, US-1.2, US-2.1, US-2.2, US-3.1, US-4.1	3+2+5+3+3+3	19	Stand up the core loop: scan a pallet, place it safely, record its location and log the movement, then find it again.
Sprint 2	US-1.3, US-2.3, US-3.2, US-4.2, US-5.1, US-5.2	2+3+2+8+5+2	22	Harden the system: error handling, movement history, fast lookup at scale, and role-based access.

Backlog summary: 5 Epics, 12 User Stories, 41 story points total.

Reflection

1. Did your estimations reflect the actual effort?

Mostly, yes. The smaller scan, record and search stories (2-3 points) were quick and matched their estimates, while the two heaviest stories - rack-capacity validation (5) and the sub-100 ms lookup across 100,000 bins (8) - held the real uncertainty, which is why they carried the highest Fibonacci values. In hindsight US-4.2 could have been split, since performance work is genuinely hard to size before you attempt it.

2. Was your backlog well-prioritized?

Yes. Safety-critical and core-flow stories (identify, place, validate rack capacity, record the movement and search) were pulled into Sprint 1, because nothing else matters if a rack can be overloaded or a pallet cannot be located. Error handling, audit history, performance hardening and role-based access followed in Sprint 2 once the core loop worked end to end.

3. How did your simulated sprint align with your plan?

Closely. Sprint 1 committed 19 points and Sprint 2 committed 22 points, and in both sprints all six stories moved To Do -> In Progress -> Done inside the one-week box. Scope stayed stable and nothing carried over, so the plan and the executed sprints matched.

4. What insights did the burndown chart give about your team's capacity?

Sprint 2's remaining-work line tracked close to the ideal guideline, which shows the 22-point commitment was realistic and well paced. Sprint 1's chart, by contrast, dropped almost vertically because the work was completed in a single working session rather than spread across the week - a reminder that in a real team, completions should be paced daily so the burndown reveals true velocity. Overall, a steady 19-22 points per week looks like a sustainable capacity for this team.

Graded deliverables: Jira screenshots (backlog with Epics + Stories showing story points, the active sprint board, and both sprint burndown charts) plus this reflection document.